

④ Comparing two helpdesk structures.

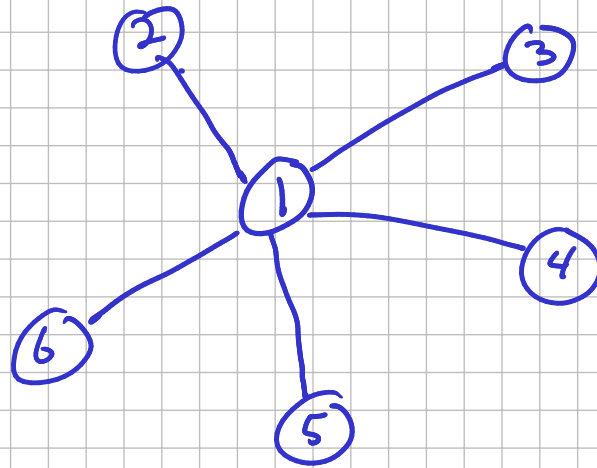
A company compares 2 possible escalation networks / on 6 agents.

Network A:

1 - 2 - 3 - 4 - 5 - 6

Network B: star topology with node 1 at the center.

$$E = \{ (1,2), (1,3), (1,4), (1,5), (1,6) \}$$
$$V = \{ 1, 2, 3, 4, 5, 6 \}$$



a) Compute the APL of network A.

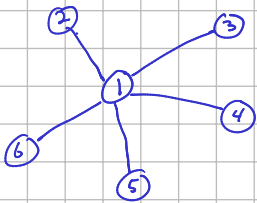
$$APL = \frac{1}{N(N-1)} \sum_{i,j=1}^N d_{i,j}$$

$$APL = \frac{1}{6(6-1)} \cdot \left[(1+2+3+4+5) \Big|_{i=1} \right. \\ \left. + (1+1+2+3+4) \Big|_{i=2} \right. \\ \left. + (2+1+1+2+3) \Big|_{i=3} \right. \\ \left. + (3+2+1+1+2) \Big|_{i=4} \right]$$

$$\begin{aligned}
& + (4 + 3 + 2 + 1 + 1) \Big|_{i=5} \\
& + (5 + 4 + 3 + 2 + 1) \Big|_{i=6} \Big] \\
& = \frac{1}{30} [(5) + (11) + (9) + (9) + (11) + (15)] \\
& \quad \frac{1}{30} (70) = \frac{35}{15} = \underline{\underline{\frac{7}{3}}}
\end{aligned}$$

b) APL for Network B:

$\Rightarrow$ Star network means each node is one step from the center and 2 steps from each other.



$$\begin{aligned}
APL = \frac{1}{30} \cdot & \left[(1 + 1 + 1 + 1 + 1) \Big|_{i=1} \right. \\
& + (1 + 2 + 2 + 2 + 2) \Big|_{i=2} \\
& + (1 + 2 + 2 + 2 + 2) \Big|_{i=3} \\
& + (1 + 2 + 2 + 2 + 2) \Big|_{i=4} \\
& + (1 + 2 + 2 + 2 + 2) \Big|_{i=5} \\
& \left. + (1 + 2 + 2 + 2 + 2) \Big|_{i=6} \right]
\end{aligned}$$

$$= \frac{1}{30} [5 + 9 + 9 + 9 + 9 + 9]$$

$$= \frac{1}{30} \cdot 50 = \frac{25}{15} = \underline{\underline{\frac{5}{3}}}$$

c) Which network is better for fast information diffusion?

⇒ Network B - it has lower APL!

d) Which network creates a stronger dependency on one node?

⇒ Network B - Node 1 is a central hub

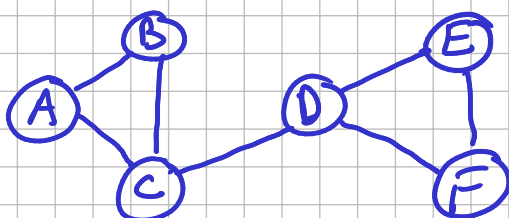
→ If Node 1 goes down, the entire network fails!

e) Explain the business trade-off between efficiency and robustness.

Pro ⇒ Higher connectivity means better speed

Con ⇒ Higher connectivity means not as robust to node failures!

⑥ A referral network among 6 customers is given by $E = \{(A,B), (A,C), (B,C), (C,D), (D,E), (D,F), (E,F)\}$



a) Compute the local clustering coefficient of each node.

$$CC_i = \frac{2 \cdot L_i}{k_i(k_i-1)}$$

$$CC_A = \frac{2 \cdot L_A}{k_A(k_A-1)} = \frac{2 \cdot 1}{2 \cdot 1} = 1 \quad \checkmark$$

$$CC_B = \frac{2 \cdot L_B}{k_B(k_B-1)} = \frac{2 \cdot 1}{2 \cdot 1} = 1 \quad \checkmark$$

$$CC_C = \frac{2 \cdot L_C}{k_C(k_C-1)} = \frac{2 \cdot 1}{3 \cdot 2} = \frac{2}{6} = \frac{1}{3} \quad \checkmark$$

$$CC_D = \frac{2 \cdot L_D}{k_D(k_D-1)} = \frac{2 \cdot 1}{3 \cdot 2} = \frac{2}{6} = \frac{1}{3} \quad \checkmark$$

$$CC_E = 1 \quad \checkmark$$

$$CC_F = 1 \quad \checkmark$$

b) Which nodes are most strongly embedded in a local group?

Info: A node is considered strongly embedded in a local group if its $CC = 1$.

$\Rightarrow A, B, E, F$

c) Which nodes act as potential bridges between two groups?

$\Rightarrow C, D$

d) Which customer would you approach first for a referral campaign that should spread across the whole network?

$\Rightarrow$ Either C or D

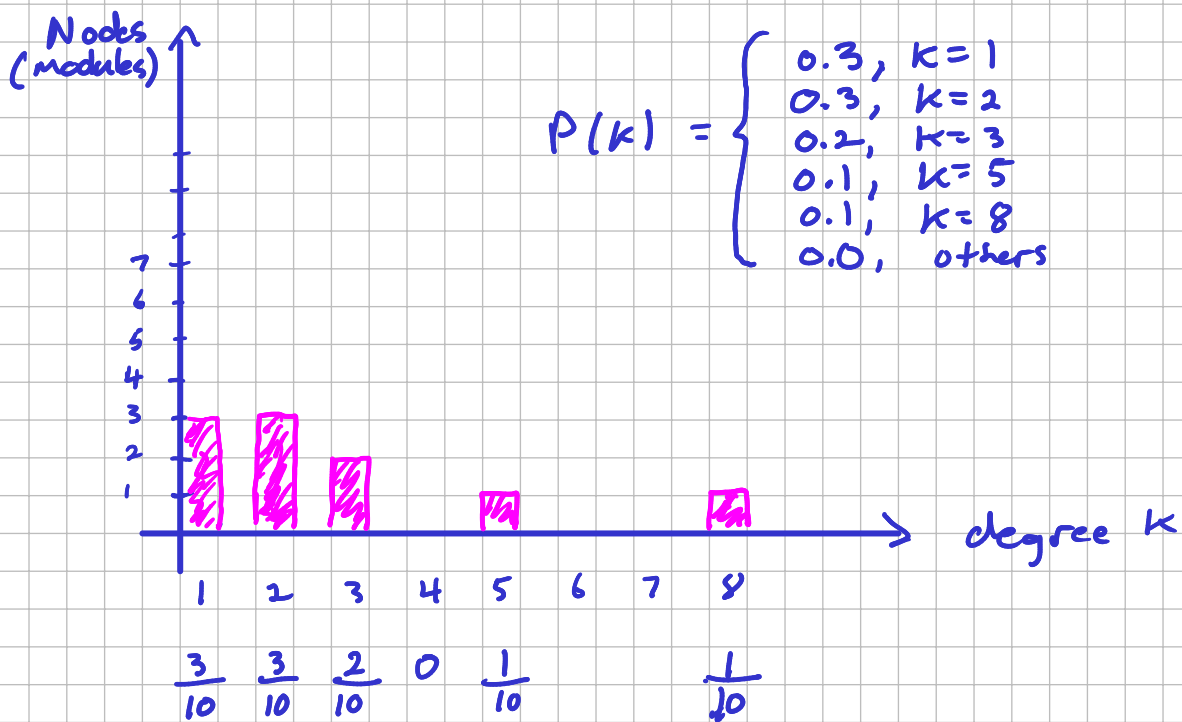
$\Rightarrow$ They are bridges in the referral network, because they are more central: referrals will disseminate faster!

⑨ Degree distribution in an enterprise application network.

The degrees of 10 enterprise software modules are

1, 1, 1, 2, 2, 2, 3, 3, 5, 8.

a) Construct the degree distribution.



b) Construct the cumulative degree distribution

⇒ How many software modules have a degree of k or more.

$$k=1 \Rightarrow P(k \geq 1) = \frac{10}{10} = 1.0$$

$$k=2 \Rightarrow P(k \geq 2) = \frac{7}{10} = 0.7$$

$$k=3 \Rightarrow P(k \geq 3) = \frac{4}{10} = 0.4$$

$$k=4 \Rightarrow P(k \geq 4) = \frac{2}{10} = 0.2$$

$$k=5 \Rightarrow P(k \geq 5) = \frac{2}{10} = 0.2$$

$$k=6 \Rightarrow P(k \geq 6) = \frac{1}{10} = 0.1$$

$$k=7 \Rightarrow P(k \geq 7) = \frac{1}{10} = 0.1$$

$$k=8 \Rightarrow P(k \geq 8) = \frac{1}{10} = 0.1$$

$$k > 8 \Rightarrow P(k > 8) = \frac{0}{10} = 0.0$$

$$P(k \geq k) = \begin{cases} 1.0, & k=1 \\ 0.7, & k=2 \\ 0.4, & k=3 \\ 0.2, & k=4 \\ 0.2, & k=5 \\ 0.1, & k=6 \\ 0.1, & k=7 \\ 0.1, & k=8 \\ 0.0, & k > 8 \end{cases}$$

c) Compute the average degree $\langle k \rangle$!

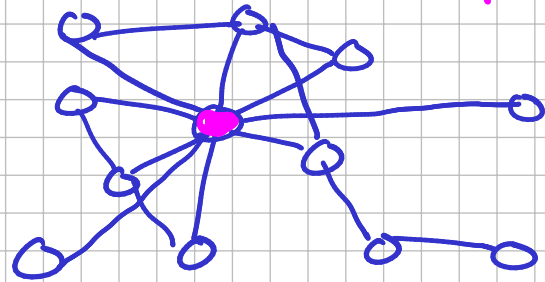
$$\langle k \rangle = \frac{1+1+1+2+2+2+3+3+5+8}{10}$$

$$\langle k \rangle = \frac{28}{10} = 2.8$$

d) Identify the hub modules.

Note: In network science, hubs are nodes in a network where

$$k \gg \langle k \rangle$$



! $\Rightarrow$ Since the node with $k=8$ connects to 80% of our network, this is a hub

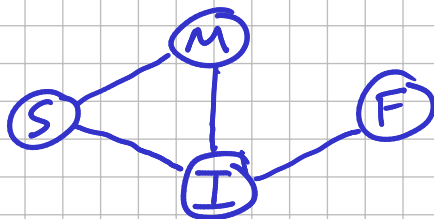
e) Explain the operational risk if a hub module fails.

$\Rightarrow$ In enterprise applications, if a highly connected module fails, and it also is a hub, then the entire network can become fragmented.

$\Rightarrow$ This means many applications with low degree might become inoperable!

⑬ Constructing the Laplacian Matrix. (ERP Integration)

$$V = \{S, M, I, F\} \quad E = \{(S, M), (S, I), (M, I), (I, F)\}$$

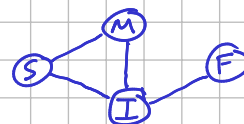


a) Write the adjacency matrix A in the order S, M, I, F .

$$A = \begin{matrix} & \begin{matrix} S & M & I & F \end{matrix} \\ \begin{matrix} S \\ M \\ I \\ F \end{matrix} & \begin{vmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{vmatrix} \end{matrix}$$

b) Write the degree matrix D

$$D = \begin{matrix} & \begin{matrix} S & M & I & F \end{matrix} \\ \begin{matrix} S \\ M \\ I \\ F \end{matrix} & \begin{vmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 1 \end{vmatrix} \end{matrix}$$



c) Compute the Laplacian matrix

$$L = D - A$$

$$L = \begin{matrix} & \begin{matrix} S & M & I & F \end{matrix} \\ \begin{matrix} S \\ M \\ I \\ F \end{matrix} & \begin{vmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 1 \end{vmatrix} \end{matrix} - \begin{matrix} & \begin{matrix} S & M & I & F \end{matrix} \\ \begin{matrix} S \\ M \\ I \\ F \end{matrix} & \begin{vmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{vmatrix} \end{matrix}$$

$$L = \begin{matrix} & \begin{matrix} S & M & I & F \end{matrix} \\ \begin{matrix} S \\ M \\ I \\ F \end{matrix} & \begin{vmatrix} 2 & -1 & -1 & 0 \\ -1 & 2 & -1 & 0 \\ -1 & -1 & 3 & -1 \\ 0 & 0 & -1 & 1 \end{vmatrix} \end{matrix}$$

d) Verify all rows sum to zero.

✓

⑮ The quadratic form of the Laplacian.

Use the ERP integration diagram from exercise ⑬.

Suppose the vector

$$\vec{x} = \overset{s}{[4} \overset{m}{4} \overset{I}{5} \overset{F}{1}]^T}$$

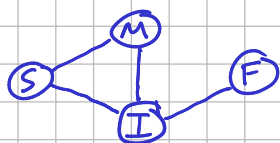
describes the digital maturity of the four systems.

a) Compute $\vec{x}^T L \vec{x}$

using the formula

$$\vec{x}^T L \vec{x} = \sum_{i,j \in E} (x_i - x_j)^2.$$

$$V = \{s, m, I, F\} \quad E = \{(s,m), (s,I), (m,I), (I,F)\}$$



$$1. (x_s - x_m)^2 = (4 - 4)^2 = 0$$

$$2. (x_s - x_I)^2 = (4 - 5)^2 = 1$$

$$3. (x_m - x_I)^2 = (4 - 5)^2 = 1$$

$$4. (x_I - x_F)^2 = (5 - 1)^2 = 16$$

$$\Rightarrow \sum_{i,j \in E} (x_i - x_j)^2 = (0 + 1 + 1 + 16) = \boxed{18}$$

verify:

$$\vec{x}' = [4 \ 4 \ 5 \ 1] \begin{bmatrix} 2 & -1 & -1 & 0 \\ -1 & 2 & -1 & 0 \\ -1 & -1 & 3 & -1 \\ 0 & 0 & -1 & 1 \end{bmatrix}$$

multiply + add

$$= [(8 - 4 - 5 + 0) \ (-4 + 8 - 5 + 0) \ (-4 - 4 + 15 - 1) \ (0 + 0 - 5 + 1)]$$

$$\vec{x}' = [-1 \ -1 \ 6 \ -4]^T$$

$$\vec{x}'^T = \vec{x}^T L$$

$$\begin{aligned}
 \vec{x}^T L \vec{x} &= \vec{x}'^T \vec{x} \\
 &= [-1 \quad -1 \quad 6 \quad -4] \begin{bmatrix} 4 \\ 4 \\ 5 \\ 1 \end{bmatrix} \\
 &= -1(4) + (-1)(4) + 6(5) + (-4)(1) \\
 &= -4 - 4 + 30 - 4 \\
 &= 18
 \end{aligned}$$

b) Explain what a large value of $\vec{x}^T L \vec{x}$ means in terms of heterogeneity between connected systems.

$\Rightarrow$ A high value for $\vec{x}^T L \vec{x}$ means higher dissimilarity between nodes (system components) in the context of maturity.

c) Explain why $\vec{x}^T L \vec{x} \geq 0$ must hold.

$\Rightarrow$ The quadratic form cannot be < 0

$$\frac{(x_i - x_j)^2}{2}$$